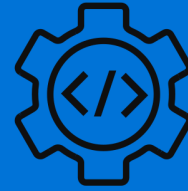


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DATA ANALYTICS CAPSTONE PROJECT - moyin.dev

Status **In progress** ▾

Timing Feb 15, 2026 to Feb 22, 2026

Owners Moyinoluwalogo Oladipo

BUSINESS CONTEXT

Owonikoko Pharmaceuticals operates three retail pharmacy branches across Lagos, Nigeria (Ikeja, Yaba, and Lekki). You have been provided with 6 months of actual sales transaction data from January to June 2026. Your mission is to analyze this data and provide strategic insights to improve business performance.

1. PROBLEM STATEMENT

Owonikoko Pharmaceuticals needs data-driven insights to optimize operations and increase profitability. Management has identified several critical business questions that require immediate attention:

Key Business Questions

1. What is the total revenue generated across all branches during the 6-month period?
2. Which products are the top performers in terms of both quantity sold and revenue generated?

3. Which product categories drive the most revenue, and are there underperforming categories?
4. How does monthly revenue trend over time? Are there any noticeable seasonal patterns?
5. Which branch is the most profitable, and how do branches compare in terms of transaction volume vs. revenue?
6. What payment methods are customers using, and does this vary by branch or customer type?
7. Are there any data quality issues that need to be addressed?

Business Impact

- Your analysis will directly influence:
- Inventory management and stock procurement decisions
- Marketing and promotional strategies
- Branch performance evaluation and resource allocation
- Revenue forecasting and financial planning
- Customer experience improvements

2. EXPECTED SOLUTION

Your analysis should provide comprehensive, actionable insights backed by data visualizations and statistical evidence.

Required Analytics Components

2.1 Revenue Analysis

- Calculate total revenue and identify trends
- Month-over-month growth analysis
- Revenue breakdown by branch, category, and product
- Seasonal pattern identification

2.2 Product Performance Analysis

- Top 10 products by revenue and quantity
- Product performance comparison across branches
- Underperforming products identification

2.3 Category Analysis

- Revenue contribution by category
- Category trends over time
- Seasonal variations by category (e.g., antimalarials, respiratory)

2.4 Branch Performance Analysis

- Transaction volume vs. average transaction value
- Branch revenue comparison and ranking
- Best-selling products per branch
- Payment method preferences by location

2.5 Customer & Payment Analysis

- Customer type distribution (Walk-in, Insurance, HMO)
- Payment method usage patterns
- Correlation between customer type and transaction value

3. RECOMMENDED ANALYTICAL APPROACH

Follow this structured methodology to ensure comprehensive analysis:

Step 1: Data Understanding & Exploration

- Import the dataset into your preferred tool (Excel, Python, Power BI, Tableau)
- Examine data structure: 1,000 rows, 10 columns
- Check for missing values and data quality issues

- Verify Revenue calculation: $\text{Revenue} = \text{Quantity_Sold} \times \text{Unit_Price}$
- Understand categorical variables (Branch, Product, Category, Payment Method, Customer Type)

Step 2: Data Cleaning

- Handle any missing values appropriately
- Convert Date column to proper datetime format
- Validate numerical ranges (Quantity: 1-20, Unit_Price: realistic ranges)
- Document any data quality issues found

Step 3: Feature Engineering

- Calculate average transaction value per branch
- Create price bands or quantity categories if useful

Step 4: Descriptive Analytics

- Calculate summary statistics (total, mean, median, min, max)
- Generate frequency distributions for categorical variables

Step 5: Visualization & Pattern Discovery

- Line charts for monthly revenue trends
- Bar charts for product and category comparisons
- Pie charts for distribution analysis
- Heatmaps for correlation analysis

Step 6: Insight Generation

- Identify trends, patterns, and anomalies
- Connect findings to business questions

4. TIPS & BEST PRACTICES

Category	Recommendations
Data Quality	Always verify revenue calculations match $\text{Quantity} \times \text{Unit_Price}$. Document any missing values and explain your handling approach.
Visualization	Use appropriate chart types for each analysis. Keep visualizations clean and labeled. Use consistent color schemes throughout.
Analysis Depth	Don't just report numbers—explain what they mean. Look for WHY patterns exist (e.g., malaria season explains antimalarial sales spike). Compare metrics across different dimensions.
Business Context	Consider Nigerian pharmacy industry context. Think about practical business implications. Remember seasonal health patterns (malaria in March-May).
Tool Selection	Use Excel for quick analysis and pivot tables. Use Python/R for advanced statistical analysis. Use Power BI/Tableau for interactive dashboards.

Key Patterns to Look For

- Seasonal Health Trends: Expect higher antimalarial sales in March-May and respiratory products in January-February
- Branch Characteristics: Lekki (premium location) should show higher average transaction values; Ikeja should have highest volume
- Product Mix: High-value items like Insulin have lower volumes but significant revenue impact
- Growth Trends: Look for month-over-month revenue growth patterns

5. DELIVERABLES

Submit a comprehensive analysis package that demonstrates your analytical capabilities and business acumen.

Format: PDF or Word Document (2-4 pages)

Content Requirements:

Date	Milestone
Feb 22, 2026 12:00 AM	Executive summary with key findings (1 page)
Feb 22, 2026 12:00 AM	Answers to all 7 business questions with supporting evidence
Feb 22, 2026 12:00 AM	3-5 key visualizations with explanations
Feb 22, 2026 12:00 AM	Strategic recommendations (minimum 5)

Optional Bonus Deliverable: Code Script

Format: Python (.py/.ipynb)

Content:

- Well-commented code showing your analysis process

Evaluation Criteria

Criteria	Weight	What We're Looking For
Data Analysis Quality	30%	Depth, accuracy, appropriate techniques
Insights & Business Value	30%	Actionable recommendations, business understanding
Visualization & Communication	20%	Clear charts, professional presentation
Completeness	15%	All questions answered, all deliverables submitted
Innovation & Creativity	5%	Unique approaches, advanced techniques

6. DATASET INFORMATION

File Details

- Filename: Owonikoko_Pharmaceuticals_Sales_Data.xlsx
- Rows: 1,000 transactions
- Period: January 1, 2026 - June 30, 2026 (6 months)
- Branches: 3 (Ikeja, Yaba, Lekki)

Data Dictionary

Column Name	Data Type	Description
Transaction_ID	Text	Unique identifier (TXN0001 format)
Date	Date	Transaction date (YYYY-MM-DD)
Branch_Location	Text	Branch name (Ikeja, Yaba, Lekki)
Product_Name	Text	Product name with dosage/specification
Category	Text	Product category (8 categories)
Quantity_Sold	Integer	Number of units sold (1-20)
Unit_Price	Currency	Price per unit in Nigerian Naira (₦)
Revenue	Currency	Total revenue (Quantity × Unit_Price)

Payment_Method	Text	Payment type (Cash, POS, Transfer)
Customer_Type	Text	Customer category (Walk-in, Insurance, HMO)

FINAL NOTES

Submission: Submit all deliverables to the email moyin.dev@gmail.com .

Questions: Contact moyin.dev@gmail.com for clarifications.

Good luck! We're excited to see your analytical insights and business recommendations for Owonikoko Pharmaceuticals. Remember: the goal is not just to analyze data, but to provide actionable intelligence that drives real business value.

RESOURCES

Project brief  DATA ANALYTICS CAPSTONE PROJECT - moyin

Dataset  Owonikoko_Pharmaceuticals_Sales_Data